ISHARA GLOVE · GRADUATION MVP · AMERICAN SIGN LANGUAGE

# Sign Vocabulary v2.2 (ASL)

20 signs + rest + other
One glove · 5 flex sensors · MPU-6050
9 October 2026 · v2.2 · replaces v1

App version 1 uses **American Sign Language (ASL)**. Every sign below is the standard one-handed ASL form, and each one was picked so that **our hardware can tell it apart** from the others. The avatar teaches exactly these forms, so the glove must be recorded with exactly these forms.

**One hand only**
One glove, on the signing (dominant) hand. No sign may need the second hand.

**No finger-to-finger touch sensing**
A sign may include a touch (thumb on fingers, hand on chest), but its meaning must not depend on detecting it.

**Different by bend, direction or motion**
Each sign differs from the rest in which fingers bend (flex sensors) or how the hand points and moves (MPU-6050).

## The 20 signs

| # | WORD | LABEL | CATEGORY | HAND SHAPE (FLEX SENSORS) | WHERE · DIRECTION · MOTION (MPU-6050) | MAY BE CONFUSED WITH |
|---|---|---|---|---|---|---|
| 1 | **One** | `one` | NUMBERS | Index straight, others closed, thumb over them | Index up, palm toward you, still | Me, You |
| 2 | **Two** | `two` | NUMBERS | Index + middle straight (V), thumb holds ring + pinky | Fingers up, palm toward you, still | No |
| 3 | **Three** | `three` | NUMBERS | **Thumb + index + middle** straight (ASL form) | Fingers up, palm toward you, still | Water |
| 4 | **Four** | `four` | NUMBERS | 4 fingers straight, thumb folded on the palm | Fingers up, palm toward you, still | Hello, Five |
| 5 | **Five** | `five` | NUMBERS | Whole hand open, fingers spread | Fingers up, still | Four, Bye |
| 6 | **Me** | `me` | PRONOUNS | Index straight | **Points to your own chest**, touches it | You, One |
| 7 | **You** | `you` | PRONOUNS | Index straight | **Points forward** at the other person | Me, One |
| 8 | **Hello** | `hello` | SOCIAL | Flat hand (B), fingers together, thumb folded | **Salute:** fingertips at the side of the forehead, hand moves out and away | Four, Thank you |
| 9 | **Thank you** | `thanks` | SOCIAL | Flat hand (B) | Fingertips at the chin, hand moves forward and down, ends palm up | Please, Hello |
| 10 | **Please** | `please` | SOCIAL | Flat hand (B), fingers together | Palm flat on the chest, circles | Sorry, Thank you |
| 11 | **Yes** | `yes` | SOCIAL | Fist (S), thumb across the fingers | Wrist nods up and down, like a head nodding | Sorry |
| 12 | **No** | `no` | SOCIAL | Index + middle straight together, **snap shut onto the thumb** | In front of the shoulder, snaps twice | Two |
| 13 | **Sorry** | `sorry` | SOCIAL | Fist (A), thumb straight up along the side | On the chest, circles | Yes, Please |
| 14 | **Water** NEW | `water` | NEEDS | **Index + middle + ring** straight (W), thumb holds the pinky | Index fingertip taps the **mouth** (lips) twice | Three |
| 15 | **Bye** NEW | `bye` | SOCIAL | Open hand, **fingers bend down and up together**, twice or more | Palm out, at shoulder height, the arm stays still | Five, Hello |
| 16 | **Come** | `come` | ACTIONS | Index straight (one-handed form) | Index points forward, palm up, then moves **toward you in one quick arc** | One, You |
| 17 | **Drink** | `drink` | NEEDS | C shape, as if holding a cup | At the mouth, tilts toward you | Hungry |
| 18 | **Call / Phone** | `call` | ACTIONS | Thumb + pinky straight (Y) | Beside the ear and cheek | I love you |
| 19 | **I love you** | `iloveyou` | FEELINGS | Thumb + index + pinky straight (ILY) | Palm out, **held still** in front (no push) | Call |
| 20 | **Hungry** | `hungry` | NEEDS | C shape | On the upper chest, moves down once | Drink |
| + | **No sign** | `rest` | EXTRA CLASS | Hand relaxed | Hand still or resting, not signing | — |
| + | **Everyday movement** | `other` | EXTRA CLASS | Any | Typing, phone, cup, scratching, talking with the hands | — |

Numbers 1–5 are signed with the palm toward you (ASL counting). Yes uses the S fist (thumb across the fingers); Sorry uses the A fist (thumb up along the side). The four finger sensors read both as a fist; only the thumb sensor differs, so the main difference the glove sees is the motion (nod vs. circle on the chest).

## What changed from v1

- **Good and Bad were removed.** Thumb up / thumb down is a gesture, not the ASL sign. ASL GOOD and BAD both start with a flat hand at the chin, almost the same as THANK YOU, so one glove can't tell them apart. They are replaced by **Water** ( `water` ) and **Bye** ( `bye` ).
- **Hello** is now the ASL salute from the forehead, not a side-to-side wave.
- **No** is now the ASL snap (index + middle onto the thumb), not an index finger shaking.
- **Three** is fixed to the ASL form: thumb + index + middle.
- New extra class `other` for everyday movements (AI v3).
- **v2.1:** Please is a flat B hand; Sorry is the A fist; I love you is held still.
- **v2.2:** Come and Water follow Lifeprint: Come is one quick arc of the index toward you (not a repeated beckon); Water taps the mouth. No label changed. References: `docs/ASL-REFERENCES.md` .

## Check before recording

1. **Watch each sign in an ASL video dictionary** (for example Lifeprint / ASL University or Handspeak) and film one team member doing it as our reference. Everyone records from that reference.
2. **Don't mix Three and Water.** Three = thumb + index + middle. Water = index + middle + ring, with the thumb holding the pinky.
3. **The flat-hand group** (Four, Five, Hello, Thank you, Please, Bye) shares almost the same bend and differs by place and motion. If the model keeps mixing them after training, drop Please first.
4. **Come** uses the one-handed form: one quick arc toward you. The repeated "come here" finger curl is a gesture, not the sign.
5. **The rest and other classes are essential.** Without them the model outputs words while the hand is doing nothing or doing everyday things.

## Demo sentences

| SENTENCE | SIGNS (ASL ORDER) |
|---|---|
| **I am hungry** | `me` + `hungry` |
| **I want water** | `me` + `water` |
| **Water, please** | `water` + `please` |
| **You, come** | `you` + `come` |
| **Call me** | `call` + `me` |
| **Hello, thank you** | `hello` + `thanks` |
| **Sorry, bye** | `sorry` + `bye` |
| **I love you** | `iloveyou` |

## Best demo for the committee

Same or almost the same hand shape: only direction, place or motion differs, which proves the motion sensor is doing real work.

| PAIR | DIFFERENCE |
|---|---|
| **Me / You** | Same finger, toward yourself or forward |
| **Yes / Sorry** | Fist (S / A), nod or circle on the chest |
| **Thank you / Please** | Same flat hand, from the chin forward or circling on the chest |
| **Three / Water** | Three fingers each, but different fingers (flex sensors) |

## Backup signs (ASL, one-handed)

| WORD | LABEL | WHAT MAKES IT DIFFERENT |
|---|---|---|
| **Like** | `like` | Only the middle finger bends; pulls away from the chest |
| **Where** | `where` | Index up, shakes side to side |
| **Mother** | `mother` | Open hand, thumb taps the chin |
| **Home** | `home` | Fingertips together, mouth then cheek |

## Recording plan

- **22 classes** (20 signs + `rest` + `other` ) × **20 takes each** × **at least 5 people** ≈ 2,200 takes, about 75 minutes of recording. Spread it over several days.
- Calibrate each person first: `python record.py calib ahmed`
- Then each sign: `python record.py water ahmed` , and everyday movements: `python record.py other ahmed`
- Long wear sessions with no signing: `python record.py wear ahmed 60` . They teach the model to stay quiet.
- **Test accuracy on a person who was not recorded.** That is the number to present at the defense.
- Review the confusion matrix after each training run, and replace signs that keep getting mixed up with a backup sign.